



PROJECT REPORT · DESENSITISED EDITION

AI-Supported Launch of a Micro Food-Retail Counter

From Remote Site Fragments to a Budget-Capped,
Compliance-Gated Opening Plan

Problem & Constraints · Layout & Utilities · Budget & Procurement · Compliance Gates
Operating Model · Brand & Documents · Outcomes & Limits

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Confidentiality and desensitisation

The cover icon is a generic pictogram of a shop. It is not the business's logo, storefront or rendering. Because parts of the underlying project are confidential, this edition has been desensitised: the brand name, logo, storefront images, city and address, landlord and supplier identities, unit prices, absolute cost, sales and margin figures, staff details and incentive ratios have been removed or generalised.

What remains is real but presented at a level that cannot identify the business: the problem structure, the workflow, dimensions and item counts, and cost shares expressed as percentages.

1 Background and Objectives

1.1 Problem Statement

A family planned to pilot a small cooked-food retail counter in a county-level city in southern China. The counter sits at the entrance of a shared retail hall and occupies a wall bay of 5.6 m by 3.0 m, about 16.8 m² of usable floor. The pilot had to prove that a compact, directly managed store model could be built cheaply, pass food-safety requirements, and be repeated later.

Three things made this hard. First, the decision-maker was not on site: a family member visited the hall and sent one annotated photograph and two voice notes, and the author had to turn these fragments into something a contractor could build. Second, the budget had a hard cap in the low tens of thousands of RMB, and the family wanted to spend on equipment, hygiene and extraction rather than decoration. Third, the landlord controlled the exhaust route, the electrical capacity and the drainage points, none of which had yet been confirmed in writing.

1.2 Situation at the Start

Table 1: Known and unknown facts at project start

Topic	Known	Unknown or unconfirmed
Space	Wall bay 5.6 m by 3.0 m at a hall entrance; existing dark ceiling, glass entrance, light floor tiles, timber slat wall	Whether the shared hall permits cooking odour and heat at this position
Extraction	An existing exhaust hood with a soft duct and small fan on the wall	Legal exit to the outside; airflow; grease leakage
Power	A supply exists in the hall	Three-phase availability and spare capacity (target at least 40 kW)
Water	Some water service nearby	Potable connection, drain size, grease trap and odour seal
Permit	A hot-food counter needs a food-business licence	Whether the regulator accepts the proposed layout and process
People	Family members can visit the site	Who will run the store day to day
Sales	A first month of trading produced a gross-margin signal	Revenue, cost of goods, wastage and net result

1.3 Success Criteria

Success was defined before any design work, in terms a third party could check (Table 2). None of the criteria is about profit, because profit cannot be known from a first month.

Table 2: Criteria set at the start

Criterion	Test
Within budget	Committed launch cost stays under the hard cap with a reserve, and every out-of-scope cost is listed before a lease is signed
Buildable from the specification	A contractor can mark the floor and connect utilities from the module table without further design fees
Safe to open	All 16 acceptance items pass on the day of inspection; any failure stops trading
Legally routed	Four pre-construction gates pass before equipment is ordered
Decision-ready	At least 30 days of complete data exist before a second store is considered

1.4 Objectives

The work had five objectives:

1. convert remote, unstructured site information into a buildable layout and utility specification;
2. create a budget and procurement system that prevents overspending *and* prevents unsafe savings;
3. define an acceptance regime that must pass before any food is sold;
4. produce brand assets and business documents that are consistent with one another and disclosed at the right level to the right audience;
5. state results honestly, without presenting early signals as proof.

1.5 Role and Boundaries

The author acted as analyst and coordinator: defining the specification, directing the AI tools, writing the control workbook and business documents, and reviewing every output against the site information and public rules. A family member did the site visit. The store manager, contractors and suppliers are outside the scope of this report, and no site inspection, permit decision or electrical design is claimed as the author's own; licensed trades were designated to re-check loads and connections.

1.6 Disclosure Scope

This report is a *desensitised edition*. The brand name, logo, storefront images, exact location, landlord and supplier identities, unit prices, absolute cost, sales and margin figures, staff details and incentive ratios are omitted or expressed as shares and ranges. Layout dimensions, item counts and process structure are kept because they show how the work was done without identifying the business. A fuller walkthrough can be given orally under a confidentiality agreement.

2 Solution Design

2.1 Overview

The solution is a pipeline in which each stage produces a document that the next stage and an outside professional can check (Figure 1). Nothing is ordered or built until the four gates in Section 4 have passed. The design rests on four principles, applied consistently across every sheet of the workbook.

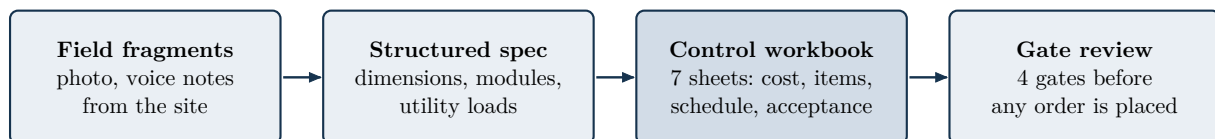


Figure 1: From site fragments to a gated build; each stage yields a checkable document

2.2 Design Principles

Spend where safety needs it. Food-contact surfaces, cold-chain and electrical items are bought new, with nameplates, certificates and invoices; decorative work is cut. **Reuse what exists.** The existing ceiling, floor tiles, timber slats and exhaust hood are kept wherever they pass inspection. **Defer upgrades.** Three items were left at zero budget until sales justify them. **Every number has a range.** Each cost line carries a low reference, a target and an alarm level, so a quote can be judged at a glance.

2.3 Why a Workbook Rather Than a Drawing

A drawing shows a shape; it does not tell a buyer what to reject. Because the family would be buying from local suppliers without a designer present, the deliverable had to carry the acceptance logic with it: for each item, the minimum parameter, the price band, the delivery test and the installation condition. The workbook therefore doubles as a purchasing brief, a site checklist and an inspection record, and each sheet refers to the next by module code (A1–A8, F1–F5, D1–D4).

2.4 Layout Logic

The layout is organised as modules along the wall and bands in depth (Figures 2 and 3). Along the 5.6 m wall there are eight back-of-house modules; in front of them five customer-side modules; and the 3.0 m depth is divided into four bands. Each of the 17 modules or bands has a start, an end and a width in millimetres, so a contractor can mark the floor directly. The staff aisle was kept at 1.2 m so that doors can open without blocking movement, and the front display was fixed at a standard 1.8 m straight unit rather than a curved or custom length.

Back of house (along the 5.6 m wall)

A1 Hand-wash	A2 Double sink	A3 Prep	A4 Cooker 1	A5 Cooker 2	A6 Soup pot	A7 Portioning	A8 Chiller
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Staff aisle

Front of house (customer side)

F1 Staff entry	F2 Till	F3 Hot-holding display (1.8 m)	F4 Packing	F5 Open / service margin
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Figure 2: Module layout along the 5.6 m wall (widths to scale, labels generalised)

Depth 3000 mm, wall to customer edge

D1 Equipment line	D2 Staff aisle	D3 Display	D4
750 mm	1200 mm	850 mm	200 mm

Figure 3: Depth bands across the 3.0 m bay (to scale)

The rule that ties the layout to the workflow is separation: raw and cooked handling are placed at different modules and separated in time, hand-washing has its own basin at the staff entrance, and the three heat sources sit under the covered length of the hood.

3 Site Specification and Utility Loads

3.1 From Fragments to Specification

The inputs were one photograph with the exhaust position circled by hand and two short voice notes giving the depth and width. An AI assistant was used to turn these into a dimensioned module table and, for each module, the electrical, water, drainage and exhaust parameters a contractor needs. The output tells the contractor to re-check every load against the final equipment nameplates, and the schedule begins with a re-measurement step; the AI output was treated as a starting specification for licensed trades, not a final drawing.

3.2 Utility Parameters

Table 3: Utility specification by system

System	Suggested parameter	Protection and test
Main power	Three-phase, at least 40 kW available, 20% spare; main switch and separate distribution board	Insulation, earthing, leakage trip and full-load temperature rise
Two cookers	One dedicated three-phase circuit each; rating from the nameplate	Enclosure rating; switch away from the heat; two-hour continuous run
Soup pot	Dedicated single-phase circuit of about 32 A	Leakage protection, earthing, water-proofing; boil and hold test
Chiller and display	Separate single-phase circuits of 10–16 A, not shared with cookers	Power-restore behaviour; four-hour empty temperature test

System	Suggested parameter	Protection and test
Water supply	Potable main with isolation valves and back-flow prevention	Pressure and full-flow leak test
Drainage	Sinks to grease trap to a lawful foul drain; slope, water seal, inspection port	Both sinks discharging at once without back-flow
Extraction	Hood length at least 2.4 m; reference airflow 3,000–4,500 m ³ /h; grease filters and condensate tray	Hot-smoke test; existing hose and fan not accepted on appearance
Make-up air	Organised supply at 70–80% of extraction, not blowing on food	Door draught and smoke check; slight negative pressure
Lighting	At least 220 lux at work surfaces; high colour rendering; 3,000–4,000 K	Light-meter spot check; no red lamps that alter meat colour

The table shows why the four gates in Section 4 are ordered as they are: three of the nine systems (power, extraction, drainage) can make the whole site unusable and none can be judged from a photograph.

3.3 Equipment Selection Logic

The equipment list has 20 items, of which 17 are funded for opening and 3 are deferred. Table 4 groups them by function; each carries minimum parameters, three price tiers, a delivery check and an installation condition rather than a brand.

Table 4: Equipment groups and selection rules

Group	Items	Selection rule
Front of house	Hot-holding display, till and scale, packing area	Standard 1.8 m straight unit; product core temperature at or above 60 °C; easily washable glass
Heat	Two induction cookers, two stainless pots, soup pot, rice cooker	Cookers new with nameplates; one main and one back-up pot; run two hours without overheating
Cold chain	Dual-temperature upright chiller; deferred cold worktop	Chilled 0–8 °C, frozen at or below –18 °C, separated compartments, external display
Washing and hygiene	Double sink, hand-wash basin, sterilising cabinet, water heater, grease trap	Stainless food-contact surfaces; non-touch tap on the hand-wash basin; not shared with other sinks
Working surfaces	Stainless bench with splashback, food-grade trays, knives, boards, probes	Colour-coded raw and cooked; two probe thermometers
Safety and records	Fire extinguisher and blanket, temperature recorder, basic camera, pest lamp, bins	No expired second-hand safety items; reuse existing camera if possible
Deferred	Water purifier, cold worktop, third cooker	Buy only when sales and capacity justify

4 Budget, Procurement and Compliance Gates

4.1 Budget Control

The control workbook has seven sheets (Table 5). The committed launch budget was held at about 86% of the hard cap, leaving roughly 14% as a safety margin (Figure 4), and its structure by share is shown in Figure 5. Costs outside the workbook scope, namely rent, deposits, licence fees, new ducting, power upgrades and fire-system changes, are listed separately and must be confirmed before any lease is signed.



Figure 4: Committed budget against the hard cap



Figure 5: Structure of the committed launch budget (shares only, no absolute figures)

Table 5: Contents of the control workbook

Sheet	Content and scale
Budget overview	Committed budget vs. cap; 8% contingency line; costs excluded from scope (rent, deposits, licence fees, ducting, power upgrade)
Equipment list	20 items with low / target / alarm prices and delivery checks; 17 funded for opening, 3 deferred
Construction items	15 work items with prerequisites, acceptance criteria and responsible trade
Layout and utilities	17 dimensioned modules and bands; electrical, water, drainage and exhaust parameters
Schedule	8 stages from pre-checks to trial opening; payment caps and “do-not” rules per stage
Acceptance checklist	16 pass/fail items across licence, hygiene, temperature, electrics and pests
Basis and quoting	Regulatory basis; a six-point quotation template (price basis, parameters, after-sales, acceptance, lead time, documents)

4.2 Construction Scope Logic

The 15 construction items follow one rule: do only what safety, extraction and identification require. Items that were cut or reduced are as informative as those kept. The build does not remove the timber slat wall, does not make a large door frame, does not lay new tiles and does not create a wash-light system. It does repair only the wall behind the sign, add a back-lit sign, refurbish the existing extraction, add a small distribution board, connect short water and drain runs, seal equipment feet, and close pest entry points. Every quotation must separate materials, labour, equipment and tax documents.

4.3 Procurement Method

Each equipment and construction line has a low reference, a target and an alarm level, so a quote can be classified on sight. Every specification is bought as the same item from at least three suppliers using a six-point template: price basis (delivered, installed, commissioned, invoiced), technical parameters, after-sales terms, acceptance and return responsibility, lead time and deposit, and documents (certificate, nameplate, manual, test data). Public price pages were used only to judge market range and were not treated as a supply channel.

4.4 Compliance Gates

Four gates must pass before equipment is ordered (Table 6). Their purpose is to stop the most expensive failure mode: buying equipment for a site whose exhaust, power or water cannot support it. The workbook states that if any gate fails, ordering stops and the budget is recomputed. Design choices were grounded in public regulation on food-business licensing, catering food-safety operation and the national catering hygiene standard, cited in the workbook so that an inspector or contractor can follow the source.

Table 6: Pre-construction gates

Gate	Requirement before ordering equipment
1 Permit pre-check	Take the floor plan, menu and process to the local regulator to confirm the permitted scope and layout
2 Exhaust route	Landlord confirms in writing that the existing duct exits legally to the outside; hot smoke test with all cookers running
3 Power capacity	Confirm three-phase supply with spare capacity; if unavailable, compare an upgrade against a change of process before buying cookers
4 Water and drainage	Confirm potable supply, drain connection, grease trap and odour seal; never connect to rainwater pipes

5 Schedule, Acceptance and Food-Safety Discipline

5.1 Stage Plan

The schedule has eight stages (Figure 6) and one rule: review first, order second, enter site third. Payments are staged so that money follows verified progress: deposits are capped at 30% before drawings are frozen, progress payments at 30% before concealed works are inspected, and 5% is retained as a warranty holdback until commissioning is recorded.

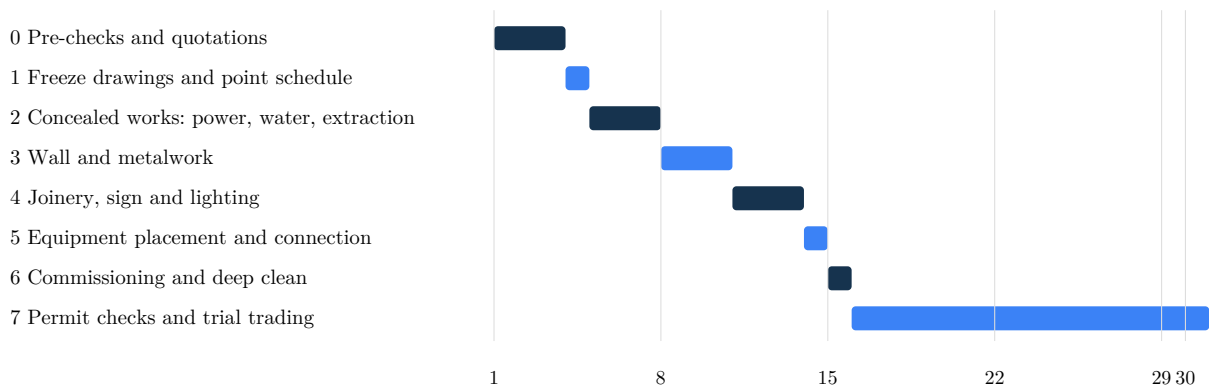


Figure 6: Eight-stage schedule (day 1 to day 30)

5.2 Opening Acceptance

Sixteen pass/fail items must be recorded before any food is sold (Table 7). Each has a judging standard, a test method and a responsible role; at the time of writing all are recorded as unchecked, because the checklist is a template for the opening inspection.

Table 7: Acceptance checklist by category

Category	Items	Example standard and method
Permit and people	Licence scope; health certificates and training	Actual products and process match the licence; certificates valid; hand-wash and temperature training signed
Extraction and utilities	Hot smoke; electrics; water and drainage	No visible smoke in the customer area with three cookers running for 30 minutes; leakage trips operate; both sinks drain at once
Temperature	Chilled and frozen; hot holding; cooking	Chilled 0–8 °C, frozen at or below –18 °C after four hours empty; hot food at or above 60 °C; cooked centre at or above 70 °C
Hygiene	Raw and cooked separation; hand-washing; cleaning and disinfection; contact surfaces; pests	Colour and text markings; non-touch tap and supplies; smooth, crack-free surfaces; sealed gaps and drain covers

Category	Items	Example standard and method
Lighting	Work-surface lux	At least 220 lux; lighting does not change food colour
Records and red line	Purchase, temperature, wastage and giveaway logs; leftover handling	Logs started on day one; food beyond its time limit or out of temperature is never sold next day

5.3 Operating Red Lines

The internal manual sets eight non-negotiable rules: any unknown-source ingredient, expired sale, falsified temperature or serious cross-contamination stops sales at once; core cooking, hot-holding, chilled and delivery temperatures have recorded standards; opening, mid-service and closing hygiene checks are done three times a day; every spoiled, dropped or out-of-temperature item is logged with name, quantity, reason and person; suppliers and meat documents allow trace-back by batch and date; all takings go through one account with daily reconciliation of sales, stock counts, giveaways and wastage; equipment models, temperatures and times are standardised; and each repair or cleaning event is recorded. Unsold food is never “given away” as a substitute for a safety judgement.

6 Operating Model, Brand and Business Documents

6.1 Operating Model

The store is directly managed. The head office controls capital, brand, recipes, supply, payment accounts, systems, training and supervision; a local clan network may recommend reliable people and help with local relationships but does not acquire equity or independent-opening rights; the store manager runs staff, hygiene, quality, stock, counting, sales and complaints but may not buy privately, change recipes, collect money outside the account or take customer data; staff learn to the standard, record data and serve. Table 8 summarises the split.

Table 8: Responsibility split in the directly managed model

Party	Responsible for	Must not
Head office	Capital, brand, recipes, supply, accounts, systems, training, supervision	Collect fees while avoiding food-safety and operating responsibility
Local network	Recommend people; help local relationships; support trust	Be treated as automatic holders of equity or opening rights
Store manager	Staff, hygiene, quality, stock, counting, sales, complaints	Buy privately, change recipes, take money outside the account, take customer data
Staff and trainees	Learn standards, follow posts, record data, serve customers	Be engaged without a labour contract, pay and exit rights

6.2 Menu and Data Discipline

The first menu is limited to 8–12 items, grouped into traffic drivers (braised meat on rice and meatball soup), signature products, and bundles of staple, soup and vegetable. Every item records daily volume, cost, price, yield, wastage and customer feedback, so the top fifth by sales and margin can be identified before the range is widened. The plan is to collect at least 30 days of complete data before any judgement on net profit or replication.

6.3 Brand and Visual Assets

An AI image tool supported the logo and a store rendering. The visual rules are compact: one primary mark, a single appearance of the name, a palette of deep red, warm grey and dark walnut with champagne gold only as thin edging, and no large bright-gold surfaces or ornate motifs. The rendering is labelled in the business document as a style reference, not a construction drawing, and the built counter was deliberately shortened relative to the picture to fit the budget. The brand name had been revised at least once; earlier versions were retired from the working folder so that only one version of each file is authoritative.

6.4 Two Business Documents and a Disclosure Rule

Two documents were written from the same facts: an internal execution manual with eleven sections and a thirty-day action list, and an external introduction with six sections. The external version contains no unit cost, margin, incentive terms, core process or supply plan. A written rule states that only the external plan may be sent before intent is clear, that visuals may be attached, and that internal figures are released only after a confidentiality arrangement.

Table 9: Disclosure levels by document

Document	Audience	Contains
External business plan	Staff, partners, first conversations	Positioning, product structure, store model, growth path, risks, partnership directions
Internal execution manual	Core team only	Decisions, cost controls, incentive design, action list, open questions
Control workbook	Core team and contractors as needed	Prices, specifications, schedule, acceptance
Site material	Verification only	Original photograph and confirmed dimensions

7 Judgement, Outcomes and Limitations

7.1 Two Judgement Calls

Labour compliance. An early incentive draft treated young staff as unpaid apprentices. The author checked it against employment law and the local minimum wage, marked it as a compliance risk, and redesigned it as base pay plus role performance plus a share of distributable profit, with the profit defined after all costs and the whole scheme subject to finance, tax and labour review before use. A ceremonial apprenticeship may remain as culture, but cannot replace a contract, pay or the freedom to leave, and wording that restricts leaving is banned.

Restraint with data. Only one early signal existed, a first-month gross margin. The documents state that it is not net profit and cannot justify replication. A second store is allowed only after three months of stable profit, no major food-safety incident, a manager able to run the store, and a stable procedure. The expansion plan is staged: prove one store, then two or three stable stores with small licensed processing, and only then consider a larger central kitchen or supply-chain investment.

7.2 Results at a Glance

Table 10: Countable results

Item	Count	Item	Count
Usable floor area	16.8 m ²	Acceptance items	16
Modules and bands with dimensions	17	Schedule stages	8
Equipment items (funded / deferred)	17 / 3	Pre-construction gates	4

Item	Count	Item	Count
Construction items	15	Menu items at launch	8–12
Workbook sheets	7	Business documents	2
Utility systems specified	9	Days of data required before a decision	30+

The main value is decision quality: the plan makes explicit when the project must stop (a failed gate) and when it may repeat (three stable months). The first counter is in its validation phase; the deliverables above exist and are in use as the working basis for build and opening.

7.3 Who Did What

Table 11: Division of work

Role	Did	Did not do
Author	Wrote the specification, control workbook and both business documents; directed the AI tools; checked outputs against site information and public rules	Inspect the site; certify electrical, extraction or drainage design
AI tools	Structured the site fragments into modules and parameters; drafted lists and documents; produced logo and rendering options	Replace measurement, licensing or trade sign-off
Family member on site	Took the photograph, gave the dimensions, dealt with the landlord and suppliers locally	Author the specification
Licensed trades and regulator	Re-check loads, connect utilities, inspect and license	Engaged through the gates and acceptance stages

7.4 Limitations

No claim is made about net profit, customer flow, repeat purchase or promotion results, because complete data do not yet exist. A promotion plan was not developed to the point of execution evidence. Contractor and supplier quotations are outside this report, and the specification has not been independently audited. The AI tools are described generically because the source files do not record which product produced each item. Because images and figures are omitted, a reader cannot re-check dimensions from this document alone.

7.5 Transferable Lessons for AI Solution Work

Turn fragments into specifications and mark every place where a professional must re-check. **Gate before you spend**, in the order of the costliest failure. **Give every number a range** so that quotes can be judged mechanically. **Separate audiences** by writing more than one version of the same facts. **Refuse to over-read early data** and write the conditions for replication before the data arrive.

7.6 Next Steps

Record 30 complete days of trading; build the full monthly income statement; settle the operating entity, the store manager arrangement and the written incentive terms; and fix the recipes and operating procedures before any replication decision. On request and under a confidentiality agreement, the module table, the workbook structure with prices masked and the acceptance template can be shown on screen; the photograph, rendering and quotations are not shown.